07001012025
Hrinishka
CSE-1, IGDTUW

# Week 2: Assignment 2

### Q1. A bank wants to predict whether a loan application will be approved or rejected.

### Is this a classification or regression problem?

This is a classification problem because the output belongs to a fixed category. In this case, the loan application can either be Approved or Rejected. Since the answer is not a numerical value but a category, classification algorithms are used.

### Which algorithm among Decision Tree, KNN, and Linear Regression would be most suitable?

Among these algorithms, Decision Tree is the most suitable because it works very well for classification problems. It can make decisions based on conditions like income, age, credit score, and previous loan history. KNN can also be used, but Decision Tree is easier to interpret and visualize. Linear Regression is not suitable because it is mainly used for predicting continuous numerical values.

### Q2. Explain the difference between a feature and a label using a real-world example.

In Machine Learning, features are the input variables used to make predictions, while the label is the output or target variable that we want to predict.

For example, in predicting house prices:

- Features: Area of the house, number of bedrooms, location, and age of the house.
- Label: Price of the house.

The model learns the relationship between the features and the label and then predicts the label for new data. In simple words, features are the information given to the model, and the label is the answer we want to obtain.

### Q3. A Decision Tree predicts that a customer will purchase a product. Explain how the model reaches this prediction.

A Decision Tree makes predictions by asking a series of questions based on the features of the data. It starts from the root node and checks conditions one by one.

For example, it may first ask:

- Is the customer's age greater than 30?
- Is the customer's income high?

- Has the customer purchased products before?

Depending on the answers, the model moves through different branches until it reaches a final node called a leaf node. The leaf node gives the final prediction, such as "Customer will purchase the product."

Thus, Decision Trees make predictions using a sequence of decisions, which makes them easy to understand.

## Q4. Why is KNN considered a supervised learning algorithm? Can KNN perform classification if labels are not available?

KNN is called a supervised learning algorithm because it learns from labeled data. During training, each data point already has a known output or class label.

When a new data point is given, KNN finds the nearest neighbors and uses their labels to decide the class of the new data point.

No, KNN cannot perform classification if labels are not available. Without labels, the algorithm would not know which category the neighboring data points belong to, and therefore it would not be able to make predictions.

## Q5. Suppose a KNN model uses K = 5 and the nearest neighbors are:

**Yes, Yes, No, Yes, No**

## What will be the prediction and why?

The prediction will be Yes.

This is because KNN uses the concept of majority voting. Among the five nearest neighbors:

- Yes = 3
- No = 2

Since "Yes" appears more frequently than "No", the model predicts Yes. The class with the highest number of votes becomes the final prediction.

## Q6. Explain the equation of Linear Regression:

$y = mx + b$

$m$

$b$

## What do x, y, m, and b represent?

In this equation:

- x represents the input or independent variable.
- y represents the output or dependent variable.
- m represents the slope of the line and shows how much y changes when x changes.
- b represents the intercept, which is the value of y when x is zero.

For example, if we want to predict marks based on study hours:

- x = Study hours
- y = Marks obtained

Linear Regression draws a straight line that best fits the data and uses it to predict future values. This equation helps us understand the relationship between variables.

## Q7. What is correlation? What do +1, 0, and -1 indicate?

Correlation measures the strength and direction of the relationship between two variables. Its value lies between -1 and +1.

+1

A value of +1 indicates a perfect positive correlation. This means both variables increase together. For example, as study hours increase, marks may also increase.

0

A value of 0 indicates that there is no relationship between the variables. Changes in one variable do not affect the other.

-1

A value of -1 indicates a perfect negative correlation. This means when one variable increases, the other decreases. For example, as speed increases, travel time decreases.

Correlation helps us understand how strongly different variables are related to each other.

## Q8. A disease detection model produced the following result:

**Actual Condition: Disease Present**

**Prediction: Healthy**

### What type of prediction is this?

This is called a False Negative.

The actual condition is that the person has the disease, but the model predicts that the person is healthy.

### Why can this be dangerous?

False negatives are very dangerous because the patient may not receive treatment at the right time. As a result, the disease can become more severe and may spread to others. In healthcare applications, reducing false negatives is very important because people's lives may depend on the prediction.

## Q9. What is the purpose of a Confusion Matrix? How is it different from a Correlation Matrix?

A Confusion Matrix is used to evaluate the performance of a classification model. It shows how many predictions are correct and how many are incorrect.

It contains:

- True Positives
- True Negatives
- False Positives
- False Negatives

Using these values, we can calculate accuracy and other performance measures.

A Correlation Matrix, on the other hand, shows the relationship between different variables in a dataset. It helps us understand whether variables are positively or negatively related.

Therefore, a Confusion Matrix is used for evaluating machine learning models, while a Correlation Matrix is used to study relationships among features.

## Q10. Compare the following algorithms.

### Decision Tree

Decision Tree works by dividing data into smaller groups using conditions. It makes predictions through a series of decisions and is easy to understand and visualize. It is commonly used for classification and regression problems.

### KNN

KNN predicts the class of a new data point by looking at the nearest neighboring points. It uses majority voting for classification and is simple to implement. It works well for smaller datasets.

### SVM

Support Vector Machine (SVM) separates different classes by finding the best boundary between them. It tries to maximize the distance between classes, which helps improve prediction accuracy. It is mainly used for classification tasks.

### Linear Regression

Linear Regression predicts continuous numerical values by finding the best straight line that fits the data. It is commonly used for predicting values like prices, salaries, and sales.

### Neural Network

Neural Networks are inspired by the human brain and consist of interconnected layers of neurons. They can learn complex patterns in data and are widely used in image recognition, speech recognition, and natural language processing. They are powerful but require more data and computational resources compared to simpler algorithms.